

Epidemiology

Pathology

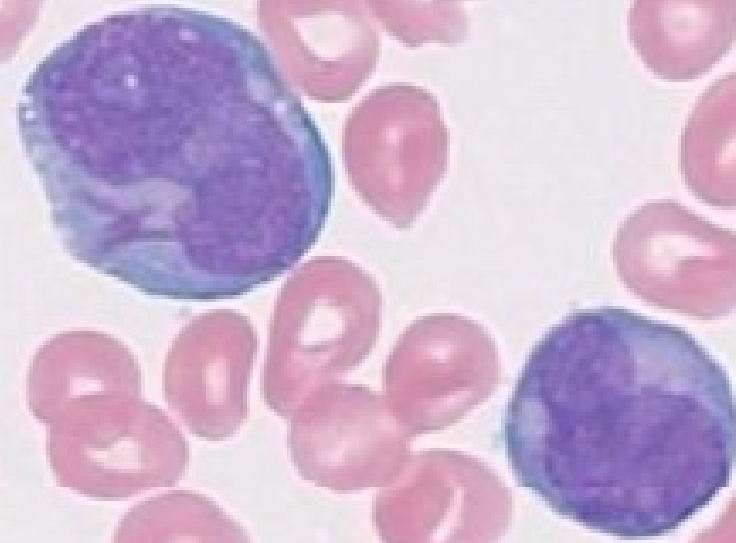
Diagnosis

Complications

Management

Cases

Acute Myeloid Leukemia



Epidemiology

Pathology

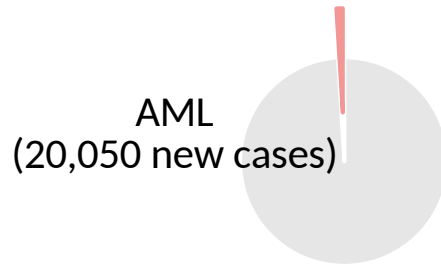
Diagnosis

Complications

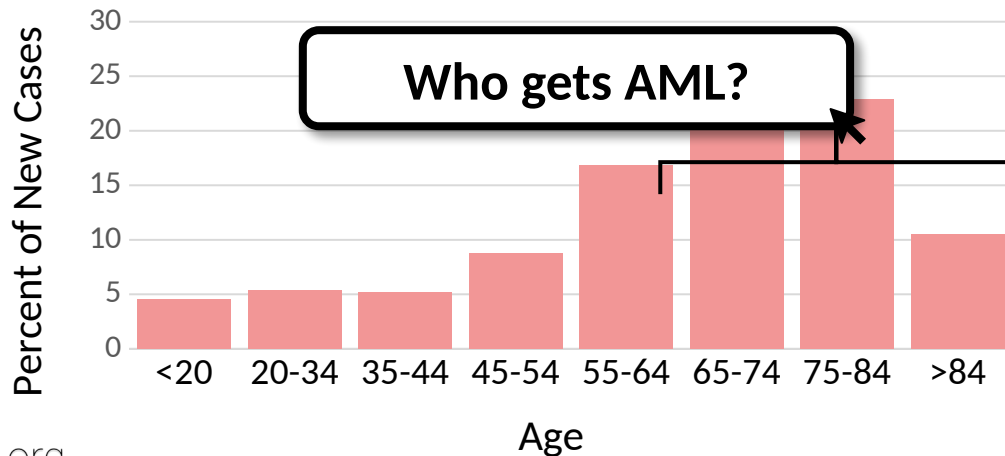
Management

Cases

AML estimated to represent
1% of all new cancer cases
in 2022

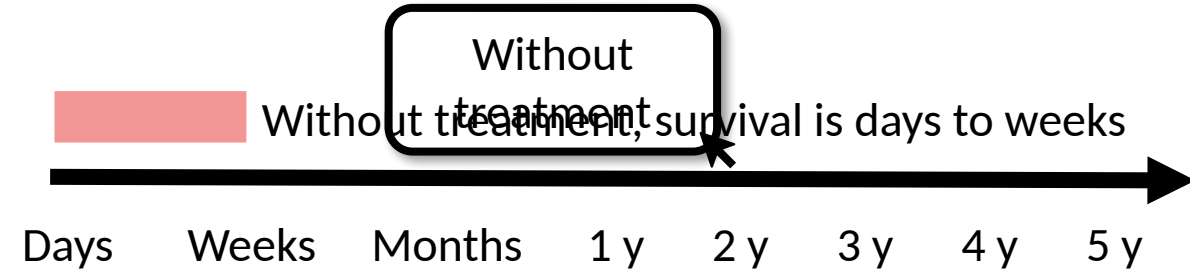


59.4% of patients diagnosed with AML are **> 65 yo**



Epidemiology

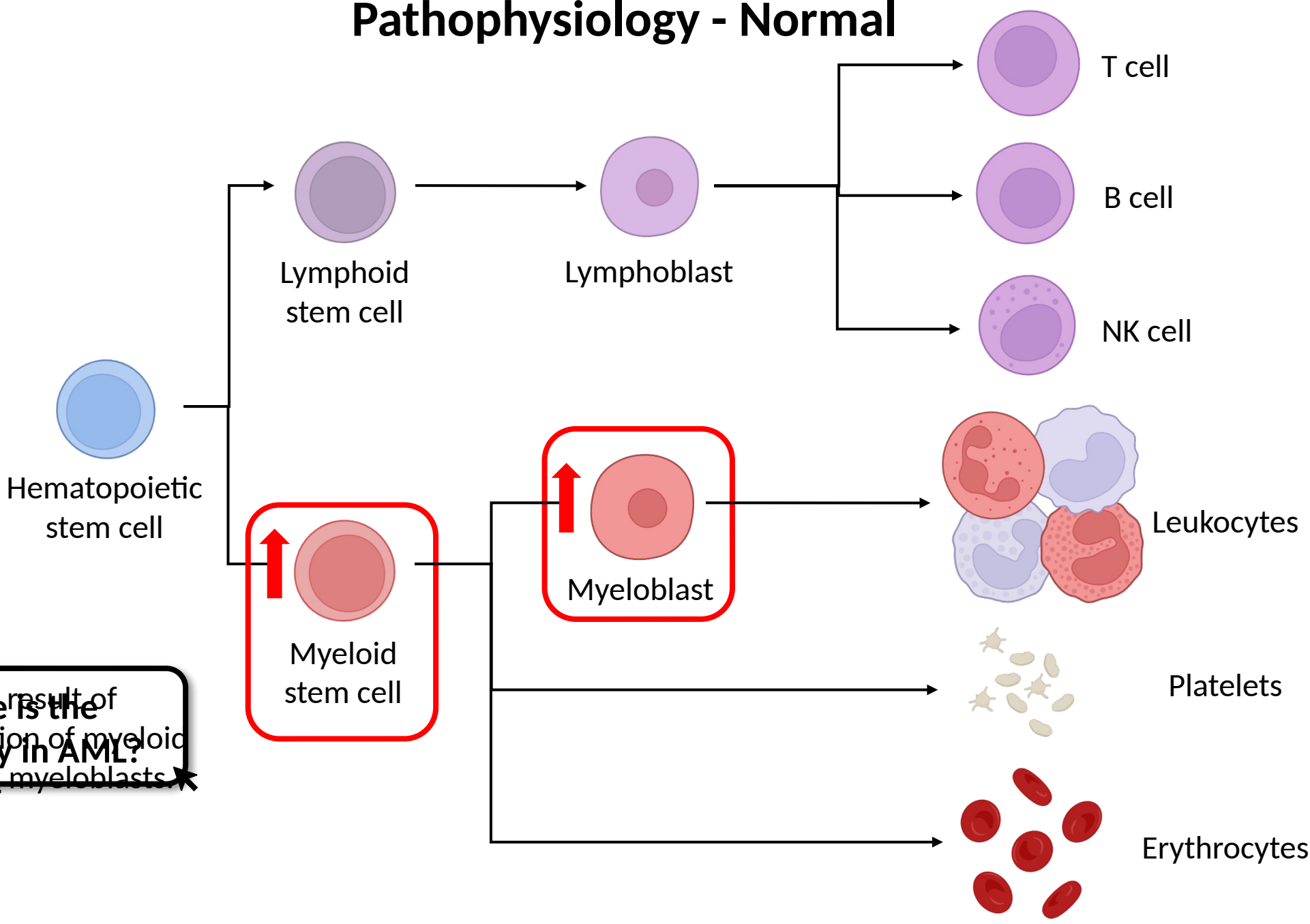
What is the prognosis?



With treatment, 5-year survival is **30.5%**

- Epidemiology
- Pathology
 - Normal
 - AML
- Diagnosis
- Complications
- Management
- Cases

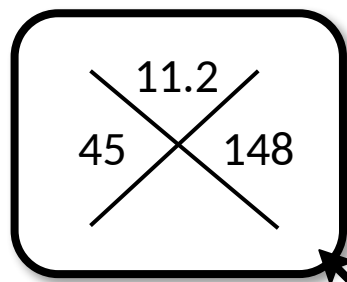
Pathophysiology - Normal



Where is the pathology in AML?
 AML is a result of overproliferation of myeloid stem cells OR myeloblasts

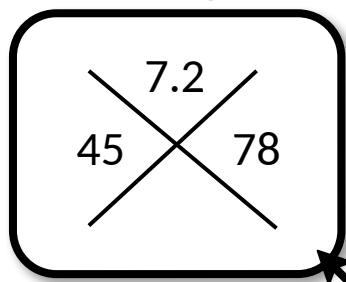
- Epidemiology
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Pathophysiology - AML



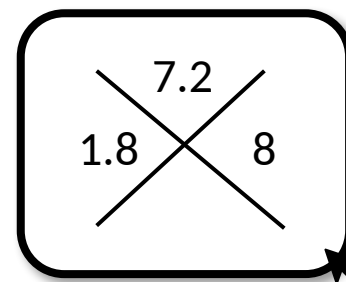
Isolated leukocytosis

↑ myeloblasts → ↑ WBCs



Leukocytosis + cytopenias

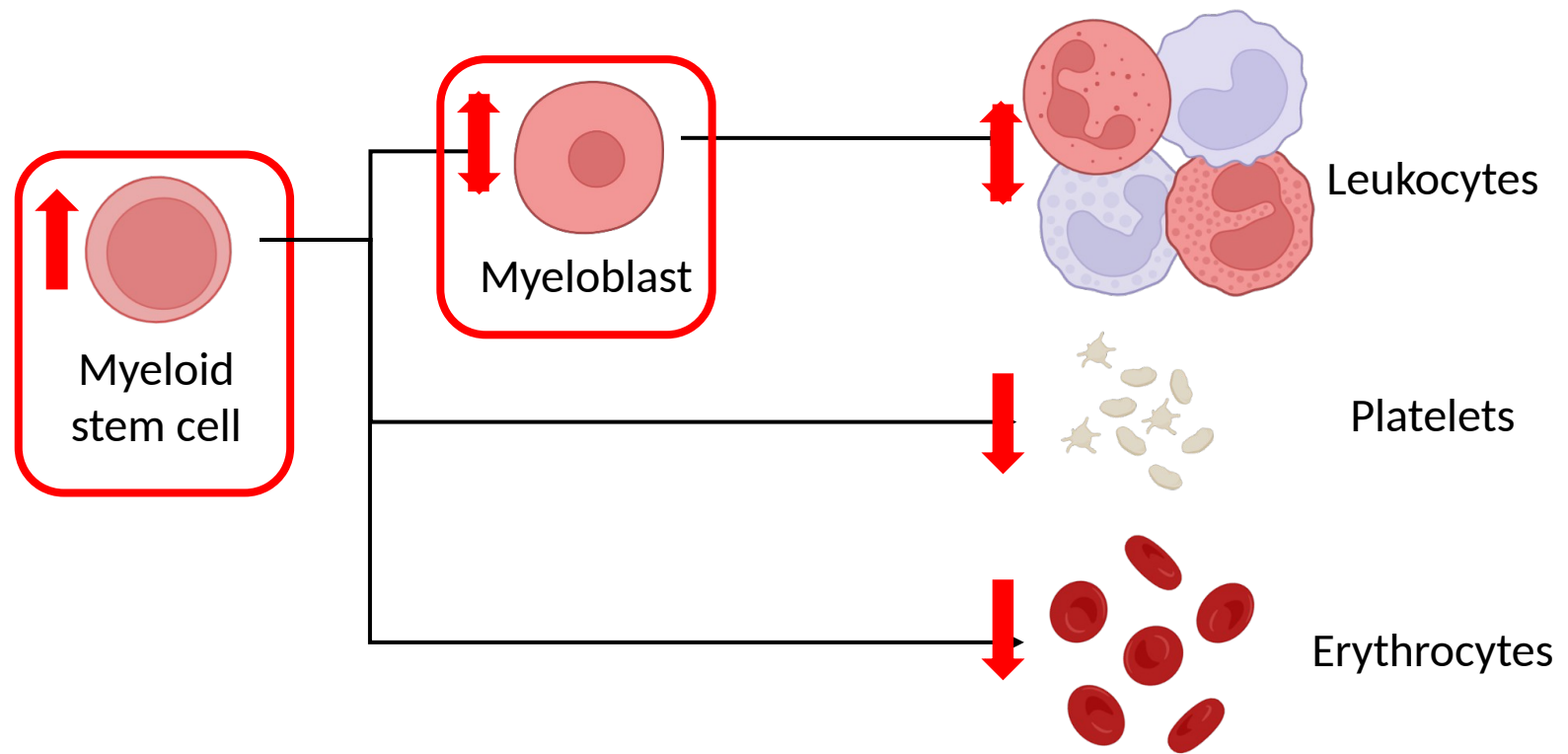
↑ myeloblasts → ↑ WBCs and displaces other cell lines in the bone marrow



Pancytopenia

↑ myeloid stem cells can displace/kill hematopoietic stems cells in the bone marrow

AML is a result of overproliferation of myeloid stem cells OR myeloblasts. It can present as a variety of different CBC abnormalities.



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Diagnosis

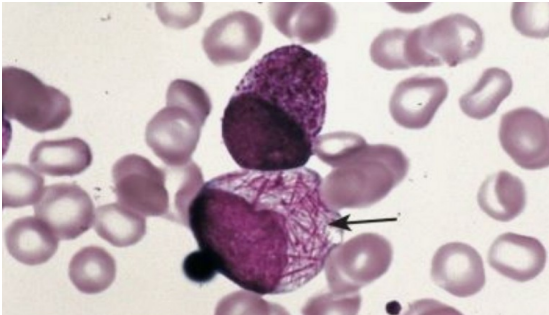
Diagnostic Criteria

1. $\geq 20\%$ blasts on *bone marrow aspirate* OR in peripheral blood
2. Cells must be of myeloid origin (Auer rods, presence of MPO, or immunophenotyping markers)

Clinical/Laboratory Findings

Differential - myeloblasts

Smear - Auer rods



Petechiae/Purpura



Gingival Hyperplasia



Leukemia cutis



Median 15k
 $>100k$ (20%)
 $<5k$ (25-40%)

$<100k$ (75%)
 $<25k$ (25%)

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Critical Complications

Headache, blurry vision, dyspnea

~~125⁹148~~

LEUKOSTASIS

↑ blood viscosity ± ↑ cytokine production by blasts

Nausea, vomiting, cardiac arrhythmias

~~135 | 6.2 | 4.8~~ ~~5.8~~
~~5.7~~

TUMOR LYSIS

Massive cell lysis with release of electrolytes and nucleic acids

Bruising, oozing from puncture sites

~~5⁷8~~ ~~77~~
~~3.5~~

DIC

Abnormal activation of coagulation and fibrinolysis

Fever, chills

~~1.8⁹148~~ ANC 500

NEUTROPENIC FEVER

↓ neutrophil-mediated inflammatory response

Signs / Symptoms

Respiratory failure ± diffuse alveolar infiltrates, fever, confusion, somnolence

Lethargy, anorexia, dysrhythmias, seizures, AKI

Bleeding, bruising, petechiae, oozing from line sites

Isolated fever, chills, malaise, myalgias, dysuria

Diagnosis

WBC > 100k OR > 50k + symptoms

>25% ↑ or value > uric acid (8) OR > 4.5
>25% ↓ in Ca

↑ coags, ↑ d-dimer, ↓ Schistocytes on smear

T > 38.3C OR > 38C x 1 h
ANC < 500 (or w/i 48 hr)

Treatment

Chemotherapy Hydration Leukapheresis

Allopurinol Aggressive IV ± Rasburicase

Induction chemo Supportive measures (transfusions)

Zosyn/Cefepime ± MRSA coverage if suggestive source

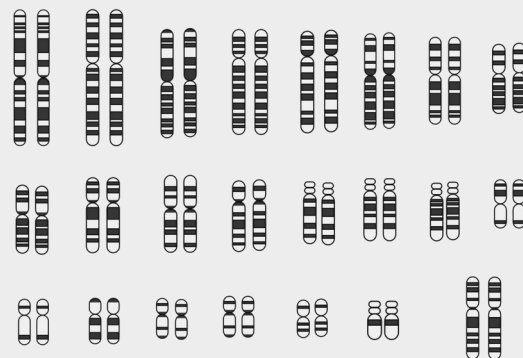
Approach to Therapy – Risk Stratification

ELN Classification

Prognosis based on

Cytogenetics

Detects large chromosomal changes such as translocations, inversions, deletions, etc



Next Generation Sequencing

Detects single gene mutations
Can guide targeted therapy

FLT3	Red	Orange	Orange	Yellow	Red	Light Green	Light Green	Green
NPM1	Green	Light Green	Light Green	Yellow	Orange	Red	Orange	Light Green
IDH1	Light Green	Yellow	Light Green	Light Green	White	Light Green	Yellow	Orange
IDH2	Orange	Red	Light Green	Green	Yellow	Yellow	Light Green	Red
NPM1	Orange	Red	Red	Orange	Yellow	Orange	Light Green	Green

Favorable

Intermediate

Adverse

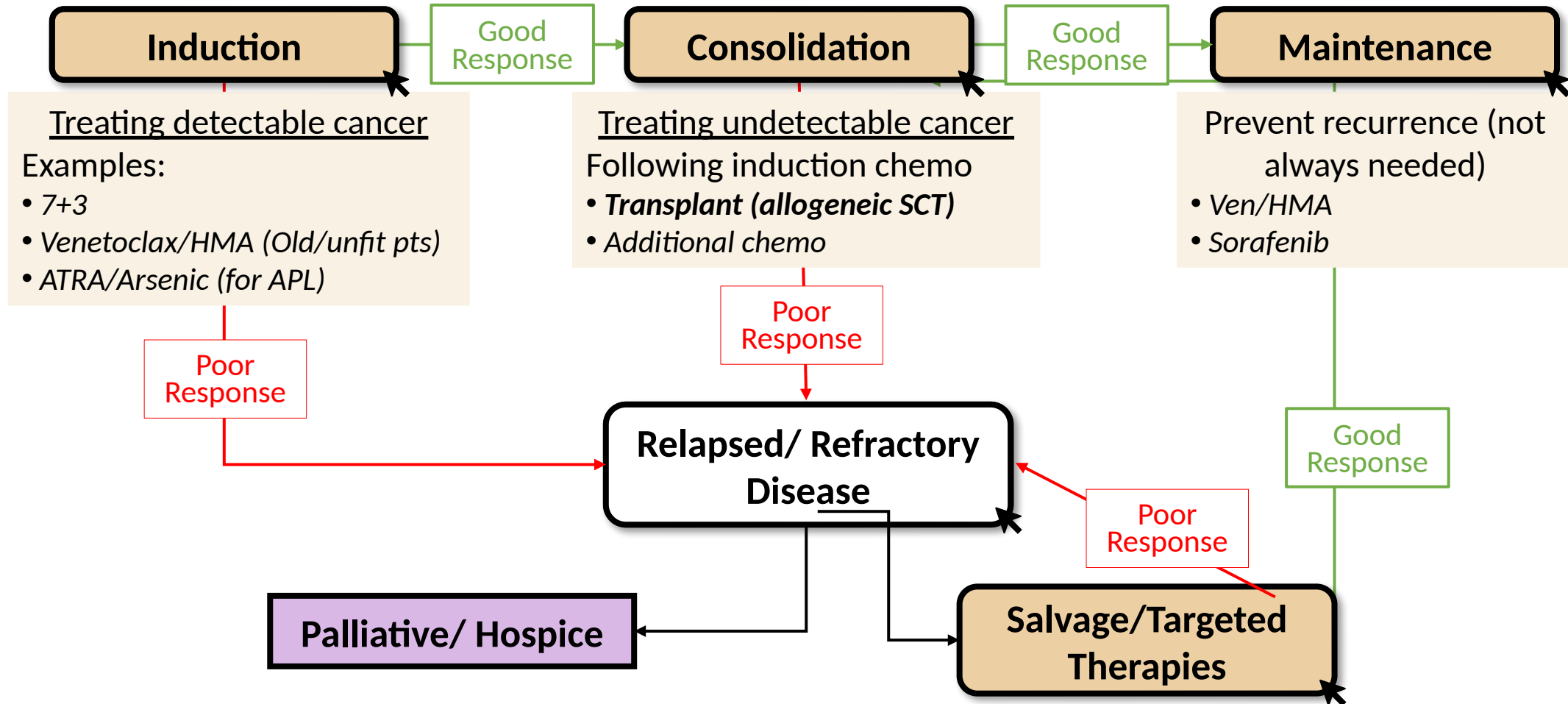
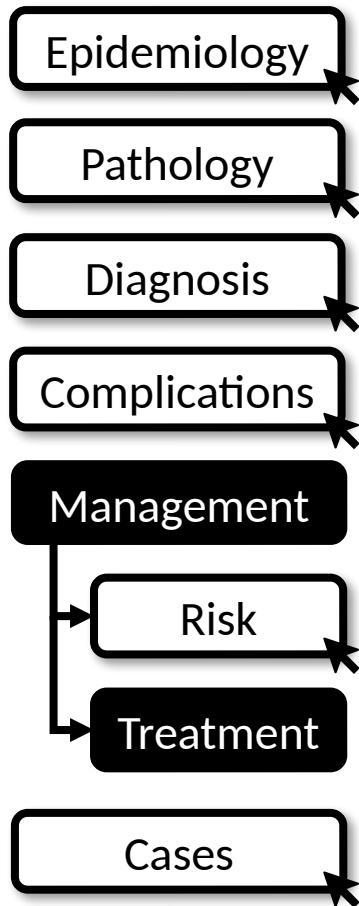
5-year OS

46%

18%

4%

Approach to Therapy - Treatment



HMA = Hypomethylating Agent (Azacitidine/Decitabine)

ATRA = All-Trans Retinoic Acid

APL = Acute Promyelocytic Leukemia

SCT = Stem Cell Transplant

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Case 1

50 year-old presents with 6 months of progressively worsening fatigue and recent malaise. On exam, T is 38.5C. CBC notable for WBC of 2.3 (absolute neutrophil count, ANC 450), Hgb 9.2, plts 72.

What is the diagnosis?

This is **neutropenic fever**, defined as a single oral T of > 38.3C or sustained > 38C for > 1h in patients with absolute neutrophil count (ANC) <500.

What is the next step in management?

Infectious evaluation (blood cultures, urinalysis, CXR, etc).
Empiric treatment is with **cefepime** or piperacillin-tazobactam. Add vancomycin if there are MRSA risk factors (CAUTI, SSTI, pneumonia, or if hypotensive).

... Smear shows myeloid blasts consistent with AML.
Cytogenetics and NGS show favorable prognosis.

How would you manage treat his AML?

Induction therapy (**e.g. 7+3 for fit patients**). Most patients with favorable prognosis likely do not require transplant.

Case 2

62 year-old presents with 6 months of progressively worsening fatigue and new gingival bleeding.

Labs: WBC 2.3 (ANC 320), Hgb 7.2, plts 16.

Peripheral smear:



What does the peripheral smear show?

Auer rod and **schistocytes**. Auer rods suggest a diagnosis of AML. Schistocytes suggest intravascular hemolysis.

What additional work-up would you want?

Coagulation panel, fibrinogen, D-dimer.

...PT/INR and PTT are elevated. D-dimer is elevated. Fibrinogen returns low.

What is the diagnosis?

Labs suggest **DIC**. Likely diagnosis is **APL**.

What is the next step in management?

Supportive transfusions. Initiate **ATRA/ Arsenic for APL**. APL has excellent prognosis with treatment.

Case 3

75 year-old presents with altered mentation, headache and blurry vision.
ROS: 30 lb unintentional weight loss

Labs: WBC 153k (23% blasts), Hgb 8.2, plt 152. BMP shows K of 6.2, Cr 1.5 (unknown baseline), Ca 6.9, Phos 5.8.

What is the likely cause of confusion and vision changes?

The clinical syndrome is concerning for **leukostasis**. Patients with AML are at risk for leukostasis with WBC > 50k. It is seen in 10-20% of new AML diagnoses.

What additional laboratory studies would you want?

Uric acid. The electrolyte derangements are concerning for possible tumor lysis, (10% of patients with leukostasis).

...Uric acid is elevated at 10.2 mg/dL.

What is the next step in management?

- Leukostasis – consider leukapheresis, start cytoreduction
- TLS – IVF, allopurinol, ± rasburicase.

Acute Myeloid Leukemia - 2022-2025 Update

What's changed since this talk (Dec 2023)

Classification (WHO 2022, ICC 2022, ELN 2022): the $\geq 20\%$ blast cutoff is NO LONGER required when an AML-defining genetic abnormality is present (e.g., PML::RARA, core-binding-factor rearrangements). NPM1-mutated AML is now a distinct entity needing $\geq 10\%$ blasts. The deck's "several nuanced exceptions" -- this is them. ELN 2022 keeps favorable/intermediate/adverse risk and adds emphasis on measurable residual disease (MRD) by NGS/flow.

Newer targeted agents reinforce the deck's point that NGS guides therapy: quizartinib (Vanflyta) added to 7+3 for newly diagnosed FLT3-ITD+ AML (QuANTUM-First; FDA Jul 2023; median OS 31.9 vs 15.1 mo) -- joins midostaurin (induction) and gilteritinib (R/R). IDH1: ivosidenib + azacitidine for newly diagnosed unfit IDH1-mut (AGILE, 2022); olutasidenib (Rezlidhia) for R/R IDH1-mut (FDA Dec 2022). Enasidenib for IDH2.

Other regimens in current use: oral azacitidine (Onureg) as maintenance after intensive induction (QUAZAR); CPX-351 (Vyxeos) for therapy-related AML / AML with myelodysplasia-related changes; gemtuzumab ozogamicin for CD33+ disease.

Standing teaching unchanged: pathophysiology (myeloid overproliferation), the CBC patterns, diagnostic criteria, the four critical complications (leukostasis, TLS, DIC, febrile neutropenia) and their management, and the induction/consolidation framework all remain current.

Acute Myeloid Leukemia

Take-Home Points

AML is a rare but high-mortality cancer of the elderly (median age 68): without treatment, survival is days to weeks; with treatment only ~1 in 3 reach 5 years.

It arises from aberrant overproliferation of myeloid stem cells / myeloblasts, which disrupts other cell lines -- expect any CBC pattern from isolated leukocytosis to pancytopenia.

Diagnosis: $\geq 20\%$ myeloid-origin blasts in marrow or blood (Auer rods, MPO+, or immunophenotype) -- OR an AML-defining genetic abnormality regardless of blast count (WHO/ICC/ELN 2022).

Know the four emergencies cold: leukostasis (WBC $>50-100k$ -> cytoreduction/leukapheresis; NOT leukapheresis in APL), TLS (allopurinol/IVF/rasburicase), DIC (low fibrinogen; treat the leukemia; suspect APL -> ATRA/arsenic), febrile neutropenia (ANC <500 -> empiric cefepime or pip-tazo).

Acute Myeloid Leukemia

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Adapted from TeachIM.org -- "Acute Myeloid Leukemia" (Hematology). Authors: Kellen Gil, MD and Maria Amaya, MD PhD; Section Editor: Chris Geiger, MD; Executive Editors: Yilin Zhang, MD and Brandon Fainstad, MD. Published December 2023.

Source URL: https://teachim.org/teaching_material/acute-myeloid-leukemia/

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Optimization (June 2026): Guideline currency verified. One deck typo fixed surgically -- "allogenic" -> "allogeneic" SCT (slide 8 treatment map); all 9 original animated/click-to-reveal slides and BioRender graphics preserved. Three slides appended: 2022-25 update, take-home, and this attribution. Newer context added (all source-cited): WHO/ICC/ELN 2022 classification (blast-threshold change, NPM1 entity, MRD); quizartinib (QuANTUM-First), ivosidenib+aza (AGILE), olutasidenib, oral azacitidine maintenance (QUAZAR), CPX-351.

Companion files: teaching guide (TeachIM-Acute-Myeloid-Leukemia-Teaching-Instructions-12.18.docx) and learner handout (AML-handout-1.pdf), both copied into the lesson folder and linked from the HTML launcher.